

Tutorial 19 Quantum Physics Suggested Solutions

- D1 (a) (i) Threshold frequency

$$f_0 = 6.4 \times 10^{14} \text{ Hz}$$

- (ii) From the graph,

$$\text{K.E.}_{\text{max}} = 4.8 \text{ eV} = 7.7 \times 10^{-19} \text{ J}$$

- (b) An electron near the metal surface absorbs one incident photon and if $f > f_0$, overcomes the attractive forces binding it to the metal. It leaves the surface with kinetic energy equal to the excess energy.

By conservation of energy,

$$\begin{aligned} \text{K.E.}_{\text{max}} &= hf - \text{min. energy to remove electron from metal surface, } \Phi \\ &= hf - hf_0 \end{aligned}$$

Rearranging,

$$h = \frac{\text{K.E.}_{\text{max}}}{f - f_0} = \frac{7.68 \times 10^{-19}}{(18.0 - 6.4) \times 10^{14}} = 6.62 \times 10^{-34} \text{ J s}$$

- D2 (a) (i) Energy of photon

$$E = \frac{hc}{\lambda} = \frac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{4.5 \times 10^{-7}} = 4.42 \times 10^{-19} \text{ J}$$

- (ii) Since $P = IA = \frac{dN_p}{dt} \times E$, the rate of incident photons

$$\frac{dN_p}{dt} = \frac{IA}{E} = \frac{(700)(1 \times 10^{-4})}{4.42 \times 10^{-19}} = 1.58 \times 10^{17} \text{ s}^{-1}$$

- (b) (i) de Broglie relation

$$\lambda = \frac{h}{p}$$

- (ii) Momentum of photon

$$p = \frac{h}{\lambda} = \frac{6.63 \times 10^{-34}}{4.5 \times 10^{-7}} = 1.47 \times 10^{-27} \text{ kg m s}^{-1}$$

- (c) (i) Change in momentum of each photon on reflection is $-2p$.

Rate of change in momentum

$$\begin{aligned} \frac{dp}{dt} &= \frac{dN_p}{dt} \times (-2p) = -(1.58 \times 10^{17})(2)(1.47 \times 10^{-27}) \\ &= -4.66 \times 10^{-10} \text{ N} \end{aligned}$$

Therefore the change in momentum in 1 second

$$\Delta p = -4.66 \times 10^{-10} \text{ N s}$$

- (ii) Photons experienced a force when they are reflected off the surface. By Newton's 3rd law, the photons exert an **equal but opposite** force on the surface. The **force per unit area** exerted on the surface is the radiation pressure.

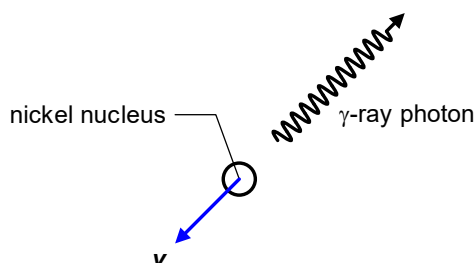
D3 (a) (i) Wavelength of γ -ray photon is

$$\lambda = \frac{hc}{E} = \frac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{(1.17 \times 10^6)(1.60 \times 10^{-19})} = 1.06 \times 10^{-12} \text{ m}$$

(ii) Using de Broglie equation, the momentum of the photon is

$$p_\gamma = \frac{h}{\lambda} = \frac{6.63 \times 10^{-34}}{1.06 \times 10^{-12}} = 6.24 \times 10^{-22} \text{ kg m s}^{-1}$$

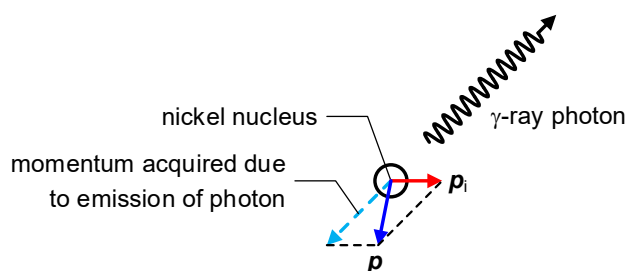
(b) (i)



(ii) By the principle of conservation of momentum,

$$mv = p_\gamma \text{ or } v = \frac{p_\gamma}{m} = \frac{6.24 \times 10^{-22}}{9.95 \times 10^{-26}} = 6.27 \times 10^3 \text{ m s}^{-1}$$

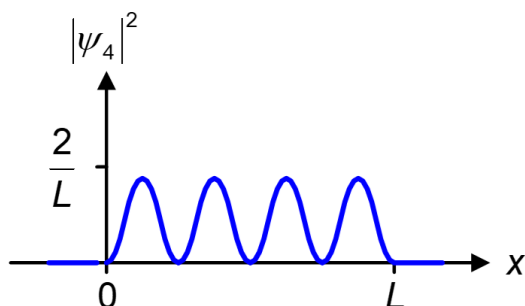
(c) The angle between the final direction of motion of the nucleus and that of the emitted photon will only be either zero or 180° if the second nickel nucleus emits photon parallel to its momentum.



If photon is emitted in other directions, then the initial momentum of the nucleus must be added to the final momentum of the nucleus determined in (b). Hence the angle between **zero** and 180° .

D4 (a) Quantum number = 4

(b)



At the nodes: 0 probability of finding the particle in δx -interval

At maxima: highest probability of finding the particle in δx -interval

(c) Mean position is $L/2$

Note:

Probability at $L/2$ is **zero**

Highest probabilities are at $L/4$, $3L/4$, $5L/4$ and $7L/4$

But because the distribution is **symmetric**,

- Equal probability on both sides of $L/2$, so average position is at the center
- Even though the particle is **never actually found there**

👉 The average value does **not** have to be a likely position.

D5 (a) A wave function that obeys $\int_{-\infty}^{\infty} |\psi|^2 dx = 1$ is said to be normalized, which implies that the particle exists at some point in space.

(b)

$$\int_{-\infty}^{\infty} |\psi|^2 dx = 1$$

$$\int_{-L/2}^{L/2} A^2 \cos^2\left(\frac{3\pi x}{L}\right) dx = 1$$

Area under the $A^2 \cos^2\left(\frac{3\pi x}{L}\right)$ graph in well of width $L = \frac{A^2}{2} \times L = 1$

$$\therefore A = \sqrt{\frac{2}{L}}$$

Recall in Chap 15 Currents, we learn how to find mean square current from a.c. sinusoidal waveform.

Or solving the integral

$$\frac{A^2}{2} \int_{-L/2}^{L/2} \cos^2\left(\frac{6\pi x}{L}\right) + 1 dx = 1 \quad (\text{use: } \cos 2\theta = 2\cos^2 \theta - 1)$$

$$\frac{A^2}{2} \left[\frac{L}{6\pi} \sin\left(\frac{6\pi x}{L}\right) + x \right]_{-L/2}^{L/2} = 1$$

$$\frac{A^2}{2} \left[0 + \frac{L}{2} - \left(0 - \frac{L}{2} \right) \right] = 1$$

$$\frac{A^2 L}{2} = 1 \quad \therefore A = \sqrt{\frac{2}{L}} \quad (\text{keep positive root for simplicity})$$

- D6 (a)** Since the ends of the string must be a node,

$$\Rightarrow a = n \left(\frac{1}{2} \lambda \right)$$

$$\Rightarrow \lambda = \frac{2a}{n}$$

Applying de Broglie equation, the momentum of the electron is

$$p = \frac{h}{\lambda} = \frac{nh}{2a}, \text{ where } n \text{ is a positive integer.}$$

(b)

$$E_n = \frac{p^2}{2m_e} = \frac{(nh/2a)^2}{2m_e} = \frac{n^2 h^2}{8m_e a^2} \text{ (shown)}$$

- D7 (a)** Photon energy, $\Delta E = E_2 - E_1 = \frac{h^2}{8mL^2} (2^2 - 1)$

$$h \frac{c}{\lambda} = \frac{3h^2}{8mL^2}$$

$$L = \sqrt{\frac{3h\lambda}{8mc}}$$

$$= \sqrt{\frac{3 \times 6.63 \times 10^{-34} (694.3 \times 10^{-9})}{8(0.13 \times 9.11 \times 10^{-31}) 3.0 \times 10^8}}$$

$$= 2.20 \times 10^{-9} \text{ m}$$

- (b)** The energy difference ΔE is inversely proportional to the square of the width: $\Delta E \propto \frac{1}{L^2}$.

As the width L decreases, the energy difference between levels increases. Because $E = h \frac{c}{\lambda}$, a higher energy transition results in a shorter wavelength of photon emitted. Light will shift towards the blue end of the spectrum.

Note this is how QLED TVs work:

1. **Backlight:** QLED TVs use a **blue LED backlight**.
2. **Quantum Dot Layer:** This blue light passes through a layer containing quantum dots of varying sizes.
3. **Color Conversion:**
 - Some dots are engineered to be large, turning the blue light into **red**.
 - Some are medium-sized, turning it into **green**.
 - Smaller dots (or just the raw backlight) pass through to provide **blue**.
4. **Result:** These pure colors are then used to create a highly accurate, bright, and vivid picture on the screen.

This, in effect, allows QLEDs to produce a more precise color spectrum than traditional LED-LCD TVs, which use filters that lose more light, resulting in lower efficiency.

- D8**
- (a)
 - (i) The lowest and most stable energy state. $n = 1$
 - (ii) The energy needed to cause an electron in an atom in a lower energy state to transit to a higher energy state.
 - (iii) The minimum energy needed to completely remove the valence electron of an atom in the ground state. $n = 1$ to $n = \infty$
 - (b)
 - (i) No transition
 - (ii) $n = 1$ to $n = 2$
 - (iii) No transition
 - (iv) $n = 1$ to $n = 3$
 - (c)
 - (i) No transition
 - (ii) $n = 1$ to $n = 2$
 - (iii) $n = 1$ to $n = 2$
 - (iv) $n = 1$ to $n = 2$ or $n = 1$ to $n = 3$

- D9**
- (a)
 - (i) Level 5 to Level 1
 - (ii) 10
 - (b) 4 spectral absorption lines.
 - (c)
 - (i) No transition
 - (ii) Level 2 to Level 1
 - (iii) Level 3 to Level 1, Level 3 to Level 2, Level 2 to Level 1
 - (d)
 - (i) Ionisation energy of sodium
 - (ii) To produce spectral line emission but no free electrons, the bombarding electrons must possess energy from 3.38×10^{-19} J to 8.21×10^{-19} J .
The corresponding accelerating p.d. is $2.11 \text{ V} \leq \text{p.d.} < 5.13 \text{ V}$.
 - (e)
 - (i) Let $\lambda_{D1} = 589.0 \text{ nm}$ and $\lambda_{D2} = 589.6 \text{ nm}$. The corresponding energy of the photons are

$$E_{D1} = \frac{hc}{\lambda_{D1}} = 3.3769 \times 10^{-19} \text{ J}; E_{D2} = \frac{hc}{\lambda_{D2}} = 3.3735 \times 10^{-19} \text{ J}$$
 So energy difference between the two levels is

$$\Delta E = 0.0034 \times 10^{-19} = 3.4 \times 10^{-22} \text{ J}$$
 - (ii) The ground state.

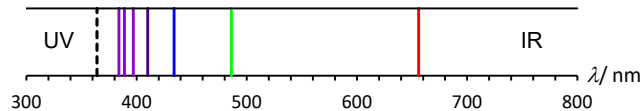
D10 (a) When $m = 4$,

$$\lambda_m = \frac{1}{R} \left(\frac{1}{2^2} - \frac{1}{4^2} \right)^{-1} = 486 \times 10^{-9} \text{ m}$$

(b) Minimum wavelength occurs when $m = \infty$.

$$\lambda_m = \frac{1}{R} \left(\frac{1}{4} - \frac{1}{\infty^2} \right)^{-1} = 365 \times 10^{-9} \text{ m}$$

(c)



(d) The probability that atoms exist in states corresponding to large m is very low. As such, transitions that produce lines with short wavelengths near 364 nm are far fewer than those involving lower m . So although there are in theory an infinite number of lines towards 364 nm (these would have formed a wide band of wavelengths), these lines are too faint to be seen and only those of longer wavelengths are visible.

D11 The de Broglie wavelength is

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}}$$

So for a given energy E , the uncertainty in position is

$$\Delta x \approx \lambda \propto \frac{1}{\sqrt{m}}$$

Since $\Delta x \Delta p \approx h = \text{constant}$,

$$\Delta p \propto \sqrt{m}$$

Therefore

$$\frac{\Delta p_p}{\Delta p_e} \approx \sqrt{\frac{m_p}{m_e}} = \sqrt{\frac{1.67 \times 10^{-27}}{9.11 \times 10^{-31}}} = 42.8$$

D12 Using de Broglie relation,

$$p = \frac{h}{\lambda}$$

we can deduced that

$$\frac{\Delta p}{p} = \frac{\Delta \lambda}{\lambda} \quad \text{or} \quad \Delta p = \frac{\Delta \lambda}{\lambda} \times p = \frac{h \Delta \lambda}{\lambda^2}$$

Hence the uncertainty in the location

$$\Delta x \Delta p \gtrsim h$$

$$\Delta x \gtrsim \frac{h}{\Delta p} = \frac{\lambda^2}{\Delta \lambda} = \frac{(300 \times 10^{-9})^2}{(300 \times 10^{-9})/10^6} = 0.300 \text{ m}$$

- D13 (a)** A particle in an infinite square well has a definite energy because the boundary conditions of the infinite square well restrict it to specific allowed energy levels. However, since the particle is confined, its position is localised and the uncertainty in its momentum is finite. According to Heisenberg's Uncertainty Principle $\Delta p \gtrsim \frac{h}{\Delta x}$, the uncertainty in the particle's momentum must be non-zero. As a result, the particle cannot have a definite momentum.

Note:

The standing wave is actually a superposition of two momentum states (+p and -p). Upon measurement of its momentum, the wavefunction collapses, and the particle is found moving in a single direction.

- (b)** If the particle has zero energy, it would be at rest and have zero uncertainty in its momentum. However, since the particle is confined, its position is localised and the uncertainty in its position is finite. This contradicts the Heisenberg Uncertainty Principle, which requires an infinite uncertainty in its position when the uncertainty in its momentum is zero. Hence, a particle confined within the box cannot have zero energy.

Since the particle must always possess some minimum non-zero momentum, it must also possess a minimum non-zero kinetic energy. Hence, the lowest energy state (known as the zero-point energy) must be greater than zero.

- C1 (a)** By having the photon source move toward the metal, the incident photons are Doppler shifted to higher frequencies, and hence, higher energy.

- (b)** For $v = 0.280c$,

$$f' = f \sqrt{\frac{1+v/c}{1-v/c}} = 9.33 \times 10^{14} \text{ Hz}$$

Therefore, work function

$$\Phi = hf_0 = 6.19 \times 10^{-19} \text{ J} = 3.87 \text{ eV}$$

- (c)** For $v = 0.900c$,

$$f' = f \sqrt{\frac{1+v/c}{1-v/c}} = 3.05 \times 10^{15} \text{ Hz}$$

So maximum K.E.

$$\text{K.E.}_{\text{max}} = hf - \Phi = 1.40 \times 10^{-18} \text{ J} = 8.77 \text{ eV}$$

C2 (a) Using time-independent Schrödinger's Equation,

$$E\psi(x) = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(x) + V(x)\psi(x)$$

$$EA(1 - x^2/L^2) = -\frac{\hbar^2}{2m} \left(-\frac{2A}{L^2} \right) + -\frac{\hbar^2 x^2}{mL^4} A$$

$$= A(1 - x^2/L^2) \frac{\hbar^2}{mL^2}$$

Hence, $E = \frac{\hbar^2}{mL^2}$.

(b) Using the normalization condition,

$$1 = \int_{-L}^L A^2 (1 - x^2/L^2)^2 dx$$

$$= 2 \int_0^L A^2 (1 - 2x^2/L^2 + x^4/L^4) dx$$

$$= 2A^2 \left[x - 2x^3/3L^2 + x^5/5L^4 \right]_0^L$$

$$= 2A^2 L [1 - 2/3 + 1/5]$$

$$= \frac{16L}{15} A^2$$

Therefore, $A = \sqrt{\frac{15}{16L}}$

(c) Probability = $\int_{-L/3}^{L/3} \frac{15}{16L} (1 - x^2/L^2)^2 dx$

$$= \frac{15}{8L} \left[x - 2x^3/3L^2 + x^5/5L^4 \right]_0^{L/3}$$

$$= \frac{15}{8} [1/3 - 2/81 + 1/1215]$$

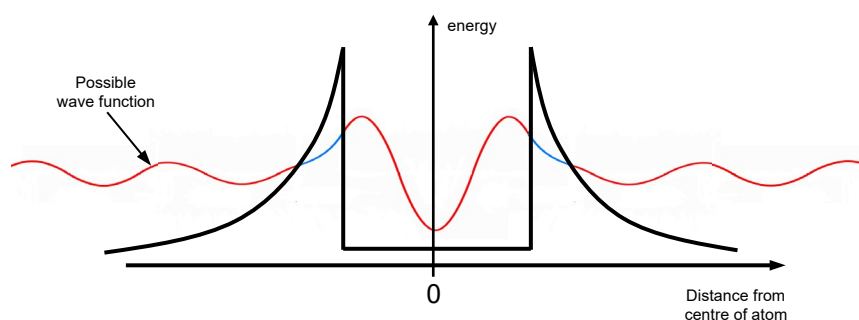
$$= 0.58$$

C3 (a) Classically, an electron could never exist outside the potential barrier imposed by the atom because it does not have sufficient energy to overcome the potential barrier. If the electron is treated as a wave, by applying Schrödinger's equation, the wave function

- is sinusoidal with large amplitude between the barrier
- decays exponentially within the barrier
- is sinusoidal with a much smaller amplitude outside the atom

Recall the square of the amplitude of the wave function gives the probability of finding the electron at a point, this means the electron has a non-zero probability of existing outside the potential well.

(b)



- C4 (a) Let the momentum of the atom be p , then

$$hf' + \frac{p^2}{2m} = \Delta E$$

- (b) By the principle of conservation of momentum,

$$p = \frac{hf'}{c}$$

So

$$\text{K.E.}_{\text{atom}} = \frac{p^2}{2m} = \frac{(hf')^2}{2mc^2}$$

Hence the required ratio is

$$\frac{\text{K.E.}_{\text{atom}}}{hf'} = \frac{hf'}{2mc^2}$$

To get an estimate of this ratio, we can consider using f instead of hf' , where $f > f'$ and $\Delta E = hf$ (energy of a photon emitted equal to the energy level difference).

$$\frac{hf}{2mc^2} = \frac{\Delta E}{2mc^2} = \frac{(1.9)(1.6 \times 10^{-19})}{2(1.67 \times 10^{-27})(3 \times 10^8)^2} = 1.0 \times 10^{-9}$$

Which shows that the ratio is very small. Hence, we can infer that using f' , which is smaller than f , it will only make the ratio even smaller. Hence, we can conclude that the effect is not very significant. (In reality, f' is only very slightly smaller than f .)

- C5 (a) For a classical atom, the centripetal acceleration is

$$a = \frac{v^2}{r} = \frac{e^2}{4\pi\epsilon_0 r^2 m_e}$$

Total energy

$$E = -\frac{e^2}{4\pi\epsilon_0 r} + \frac{1}{2}mv^2 = -\frac{e^2}{8\pi\epsilon_0 r}$$

Taking the derivative of E w.r.t. t ,

$$\frac{dE}{dt} = \frac{e^2}{8\pi\epsilon_0 r^2} \frac{dr}{dt} = -\frac{1}{6\pi\epsilon_0} \frac{e^2 a^2}{c^3} = -\frac{1}{6\pi\epsilon_0} \frac{e^2}{c^3} \left(\frac{e^2}{4\pi\epsilon_0 r^2 m_e} \right)^2$$

So

$$\frac{dr}{dt} = -\frac{e^4}{12\pi^2\epsilon_0^2 r^2 m_e^2 c^3}$$

(b) Rearranging the terms, we have

$$dt = -\frac{12\pi^2\epsilon_0^2 r^2 m_e^2 c^3}{e^4} dr$$

So time taken is

$$T = \int_{2.00 \times 10^{-10}}^0 -\frac{12\pi^2\epsilon_0^2 r^2 m_e^2 c^3}{e^4} dr = \left[-\frac{12\pi^2\epsilon_0^2 r^3 m_e^2 c^3}{3e^4} \right]_{2.00 \times 10^{-10}}^0 = 8.46 \times 10^{-10} \text{ s}$$

C6 We first make the following assumptions:

$$r \sim \frac{1}{2} \Delta x$$

$$p \sim \Delta p \approx \frac{\hbar}{2\Delta x} \approx \frac{\hbar}{r}$$

The energy of an electron near a proton is

$$U = \frac{p^2}{2m} - \frac{e^2}{4\pi\epsilon_0 r}$$

Replacing r with p ,

$$U = \frac{p^2}{2m} - \frac{e^2}{4\pi\epsilon_0 \hbar} \cdot p$$

Differentiate w.r.t. p and equating to zero to find the minimum energy,

$$\frac{dU}{dp} = \frac{p}{m} - \frac{e^2}{4\pi\epsilon_0 \hbar} = 0$$

Minimum energy occurs when

$$p = \frac{me^2}{4\pi\epsilon_0 \hbar}$$

Therefore, the energy of the electron is

$$U = \frac{me^4}{32\pi^2\epsilon_0^2 \hbar^2} - \frac{me^4}{16\pi^2\epsilon_0^2 \hbar^2} = -\frac{me^4}{32\pi^2\epsilon_0^2 \hbar^2} = -13.6 \text{ eV}$$

The radius of the hydrogen atom is

$$r \approx \frac{\hbar}{p} = \frac{4\pi\epsilon_0 \hbar^2}{me^2} = 0.53 \times 10^{-10} \text{ m}$$

So the hydrogen atom is large because if the electron is confined to a smaller space, its momentum uncertainty would increase, causing its momentum and hence energy to increase.